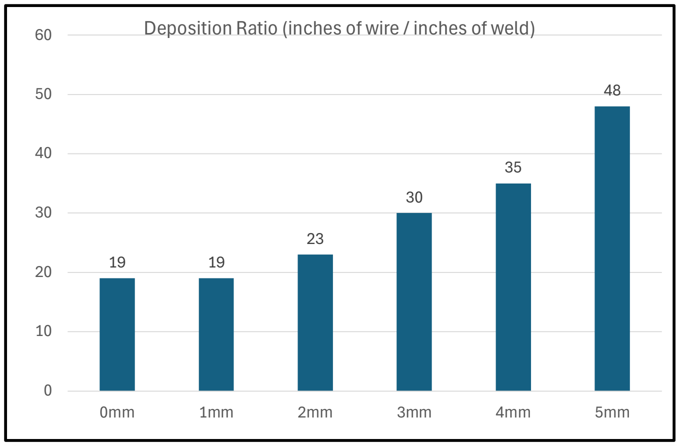
Weld Schedule Manager (Root Pass)

WPS No	July test weld 7	Position / Direction	Fixed 5G - Down	Diameter (in)	48	Wall (mm)	13.7
Process	LSC (Modified Short Arc)	**Specification**	API1104	**Filler Wire**	Surearc ER70S-6(1.1mm)	**Shield Gas**	Ar/C02 85/15
Bevel Design	30 degree	**Material Grade**	X70	**Trade Name**	Sure-arc	**Completed by**	Ben Hatch

0°	Travel Speed	Torch Angle	Dynamic Correction
0mm	14	-20	-2
1mm	12	10	-1
2mm	7	10	0
3mm	4.5	10	1
4mm	3	10	2
5mm	2	5	3

30°	Travel Speed	Torch Angle	Dynamic Correction
0mm	15	-15	-2
1mm	13	0	-1
2mm	9	0	0
3mm	6.5	0	0
4mm	4.5	-5	0
5mm	2.75	-5	0

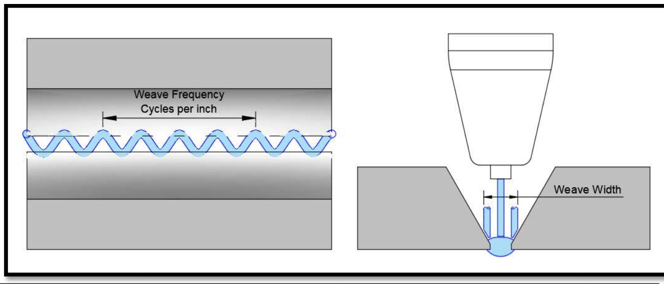


	Cycle Times (mins)	Wire Consumption LBS
0mm	2	0.3
1mm	3	0.3
2mm	4	0.4
3mm	6	0.5
4mm	9	0.6
5mm	15	0.9

Wire Diameter	1.1	
Type of line up clamp	Internal / External	
Min % root before clamp removal	75%	
Max time between bead & HP	24 Hours	
Shield Gas Flow (CFH)	55	
Min Pre-Heat Temp C° / Method	250°	Propane

60°	Travel Speed	Torch Angle	Dynamic Correction
0mm	15.5	-10	-2
1mm	13.5	-5	-1
2mm	10	-10	0
3mm	7.5	-15	0
4mm	5	-15	0
5mm	2.75	-15	0

90°	Travel Speed	Torch Angle	Dynamic Correction
0mm	16	-5	-2
1mm	14	-5	-2
2mm	10.5	-10	-2
3mm	8	-15	-1
4mm	5	-15	0
5mm	2.75	-15	0

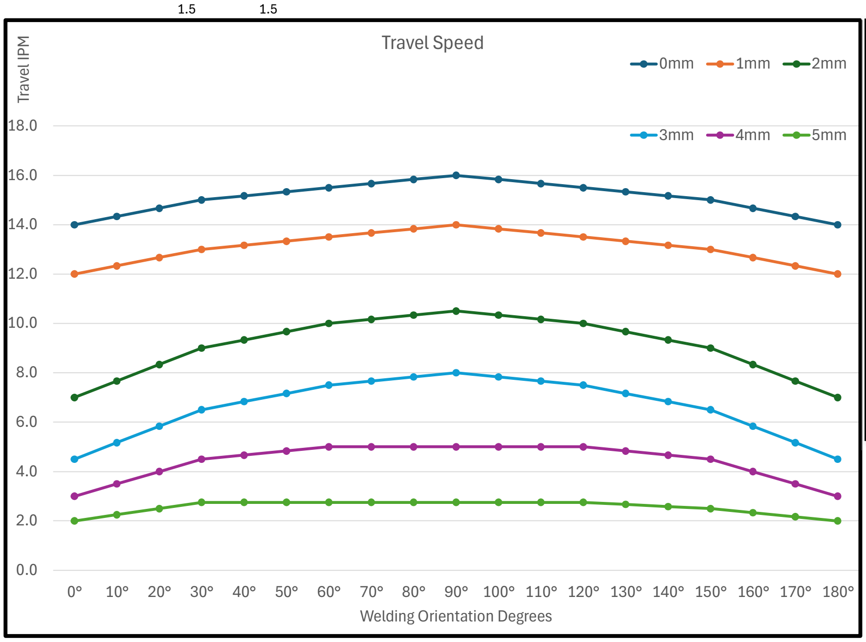


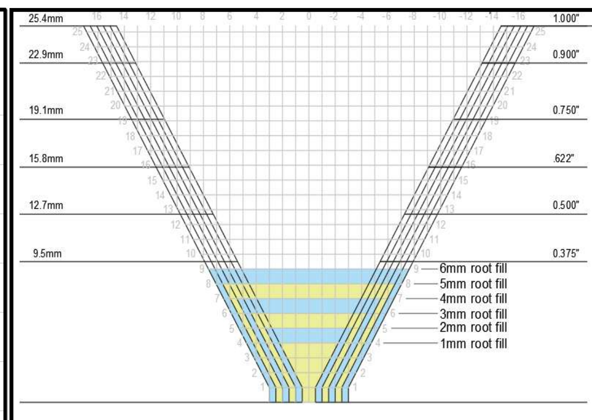
120°	Travel Speed	Torch Angle	Dynamic Correction
0mm	15.5	0	-3
1mm	13.5	-5	-3
2mm	10	-10	-2
3mm	7.5	-15	-1
4mm	5	-15	0
5mm	2.75	-15	0

180°	Travel Speed	Torch Angle	Dynamic Correction
0mm	14	0	-5
1mm	12	0	-5
2mm	7	0	-4
3mm	4.5	0	-3
4mm	3	0	0
5mm	2	0	0
		1.5	1.5

150°	Travel Speed	Torch Angle	Dynamic Correction
0mm	15	0	-4
1mm	13	0	-4
2mm	9	-5	-3
3mm	6.5	-15	-2
4mm	4.5	-15	0
5mm	2.5	-15	0

Deposition Adjustment			CTWD	Arc Length	Weave Type	Weave Frequency	Weave Width	Weave Length	Weave Dwell
Gap	Whole Pipe	Lower Half							
0mm	0%	0%	-3	0	None	0	1.2	1.2	0
1mm	0%	0%	-3	0	None	0	1.2	1.2	0
2mm	0%	0%	-2	0	Circle	12	1.4	1.4	0
3mm	0%	0%	0	0	Circle	12	3.6	3.6	0
4mm	0%	0%	1	0	Circle	12	5.2	3.8	0
5mm	20%	10%	2	0	Circle	12	6.4	4.8	0





Cleaning / Grinding				
Root Gap	Grind Weld Pass	Travel Speed	Weave Amplitude	Frequency Hz
0mm	ON	100	3	2
1mm	ON	90	3	2
2mm	ON	90	3	2
3mm	ON	90	4	2
4mm	ON	90	4.5	2
5mm	ON	90	5	2

Notes

Changes: No changes from TW6

Results: We burned through again on a point around 150 degrees. It had significant hi-low and it welded it fine after the re-start. We noticed what looked like wire restiction but was maybe from too much CTWD causing interferance with the LSC.

Weld Schedule Manager (Fill & Cap)

WPS No	July test weld 7	Position	Fixed 5G - Down	Diameter (in)	48	Wall (mm)	13.7
Process	GMAW	Specification	API1104	Filler Wire	Lincoln Pipeliner ER70-S6 (1.2mm)	Shield Gas	Ar/C02 85/15
Bevel Design	30 degree	Material Grade	X70	Trade Name	Sure-arc	Completed by	Ben Hatch

PIPE-CELL

0° Position	
Travel Speed	15
Torch Angle	0

30° Position	
Travel Speed	16
Torch Angle	-5

60° Position	
Travel Speed	18
Torch Angle	-10

90° Position	
Travel Speed	20
Torch Angle	-10

120° Position	
Travel Speed	18
Torch Angle	-10

150° Position	
Travel Speed	14
Torch Angle	-10

180° Position	
Travel Speed	10
Torch Angle	0

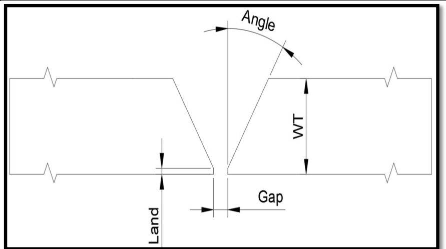
Wire Diamiter
1.2

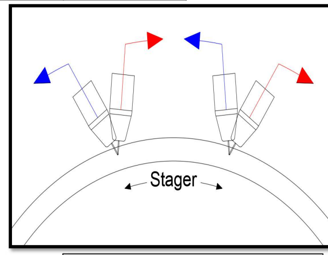
Direction of Travel
Vertical Down

Process
Pulse

Deposition Ratio
12

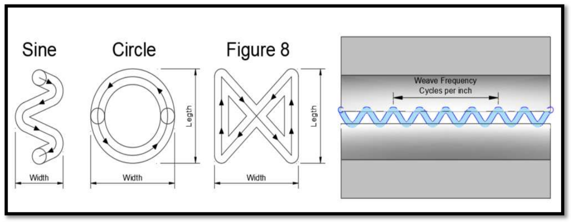
Sheild Gas Flow (L/min)
26





Joint Geometry		Tolerance (+ / -)	
Wall Thickness (mm)	13.7	1	0.5
Land (mm)	1	0.5	0.5
Root Gap (mm)	3	1	1
Bevel Angle (°)	30	5	5
Hi / Low (mm)		1.5	1.5

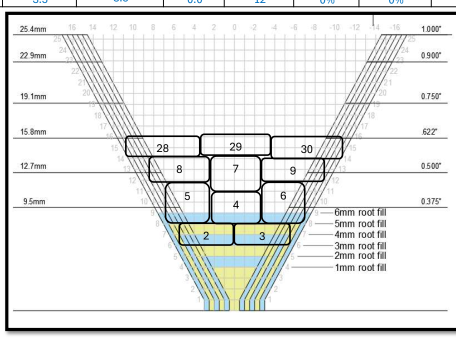
Stop / Start	
Stager the Stop-Start (mm)	30
Over run the next pass (mm)	12
Lead in at re-start (mm)	10

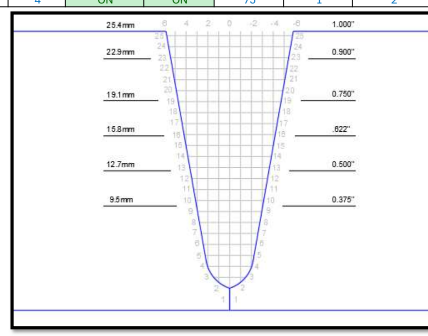


Weld Characteristics	
Deposition Efficacy	0.98
Pass Volume (mm2)	12.5
Min Cap Width (mm)	22
Root Depth (mm)	6
Total Passes (Inc root)	9

Total Arc Time (mins)	21.4
Wire Consumption (lb)	6
Wire Consumption (kg)	3

	Weave Adjustment					Weld Adjustment					Cleaning / Grinding					
	Weave Pattern	Width (mm)	Legth (mm)	Dwell (sec)	Frequency / inch	Deposition Adjustment	Travel Speed Adjustment	Pass Volume	Pass Time	Arc Legth	Grind Prior to Weld Pass	Grind After Weld Pass	Travel Speed (in/min)	Weave Amplitude	Frequency Hz	Comments
Hot Pass 0mm	Sine	1.5	2.0	0.0	12	-20%	-20%	10.6	2.9	2	OFF	ON	75	1	2	
Hot Pass 1mm	Sine	1.5	2.0	0.0	12	-10%	-10%	12.0	2.6	2	ON	OFF	75	1	2	
Hot Pass 2mm	Sine	1.5	2.0	0.0	12	-10%	-10%	12.0	2.6	2	ON	OFF	75	1	2	
Hot Pass 3mm	Sine	1.5	2.0	0.0	12	-10%	-10%	12.0	2.6	2	ON	OFF	75	1	2	
Hot Pass 4mm	Sine	1.5	2.0	0.0	12	0%	0%	13.3	2.3	2	ON	OFF	75	1	2	
Hot Pass 5mm	Sine	2.0	2.0	0.0	12	0%	0%	13.3	2.3	2	ON	OFF	75	1	2	
Fill Passes	Sine	2.0	2.0	0.0	12	0%	0%	13.3	2.3	3	ON	OFF	75	1	2	
Cap Passes	Sine	3.5	3.5	0.0	12	0%	0%	13.3	2.3	4	ON	ON	75	1	2	





			0mm		1mm		2mm		3mm		4mm		5mm		
Pass #	On/Off	Pass Type	Vert offset	Horiz	Vert offset	Horiz Offset	Vert offset	Horiz Offset	Vert offset	Horiz Offse	Vert offset	Horiz Offset	Vert offset	Horiz Offset	Comments
2	ON	Hot Pass	4.0	2.0	5.0	2.5	5.0	3.0	5.0	3.0	5.0	3.5	6.0	4.0	
3	ON	Hot Pass	4.0	-2.0	5.0	-2.5	5.0	-3.0	5.0	-3.0	5.0	-3.5	6.0	-4.0	
4	ON	Fill Pass	6.0	0.0	8.0	0.0	8.0	0.0	8.0	0.0	8.0	0.0	9.0	0.0	
5	ON	Fill Pass	6.0	3.5	8.0	4.0	8.0	4.5	8.0	4.5	8.0	5.0	9.0	5.5	
6	ON	Fill Pass	6.0	-3.5	8.0	-4.0	8.0	-4.5	8.0	-4.5	8.0	-5.0	9.0	-5.5	
7	ON	Fill Pass	8.0	0.0	9.0	0.0	9.0	0.0	9.0	0.0	9.0	0.0	10.0	0.0	
8	ON	Fill Pass	10.0	4.5	11.0	5.0	11.0	5.5	11.0	5.5	11.0	6.0	12.0	6.5	
9	ON	Fill Pass	12.0	-4.5	13.0	-5.0	13.0	-5.5	13.0	-5.5	13.0	-6.0	14.0	-6.5	
10	OFF	Fill Pass	12.0	0.0	13.0	0.0	13.0	0.0	13.0	0.0	13.0	0.0	14.0	0.0	
11	OFF														
12	OFF														
13	OFF														
14	OFF														
15	OFF														
16	OFF														
17	OFF														
18	OFF														
19	OFF														
20	OFF														
21	OFF														
22	OFF														
23	OFF														
24	OFF														
25	OFF														
26	OFF														
27.0	OFF														
28.0	ON	Cap Pass	13.0	0.0	13.0	0.0	13.0	0.0	13.0	0.0	13.0	0.0	13.0	0.0	
29.0	ON	Cap Pass	13.0	7.0	13.0	7.5	13.0	8.0	13.0	8.0	13.0	8.5	13.0	9.0	
30.0	ON	Cap Pass	13.0	-7.0	13.0	-7.5	13.0	-8.0	13.0	-8.0	13.0	-8.5	13.0	-9.0	

Notes

Changes: We made the bottom weld with no torch angle.

Results: New Baseline. It was very slightly flatter across the passes from not dragging.

Suggestions: